Project Scheduling & Resource Management

Agenda

Exercise 4: Unconstrained vs. Resource-Constrained Scheduling

Exercise 7: Parallel Method with Heuristics (Tractor Scheduling)

Exercise 10: Resource Reduction & Schedule Impact (Carpenter Scheduling)

Case Study 8.2: Power Train Ltd. Scheduling Dilemma

Case Study A8.1: The CCPM Dilemma Report

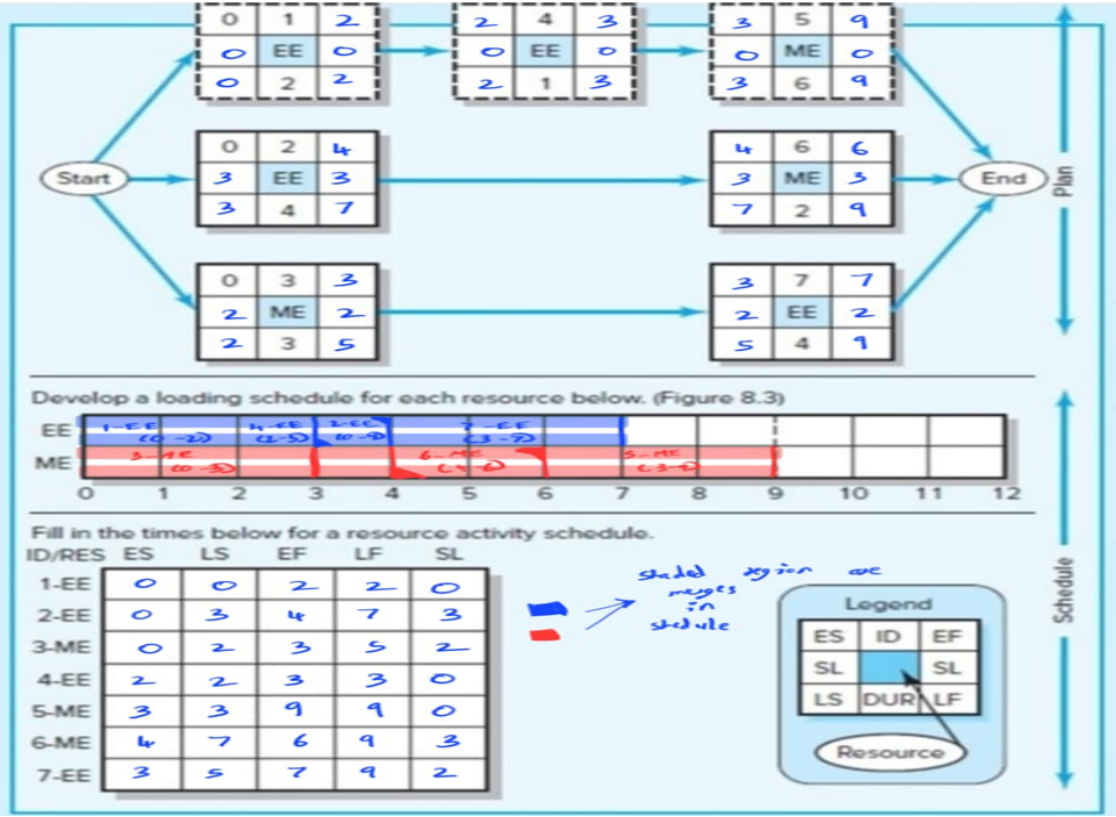
Exercise 4 - Unconstrained Schedule

Network Plan Analysis (Unlimited Resources)

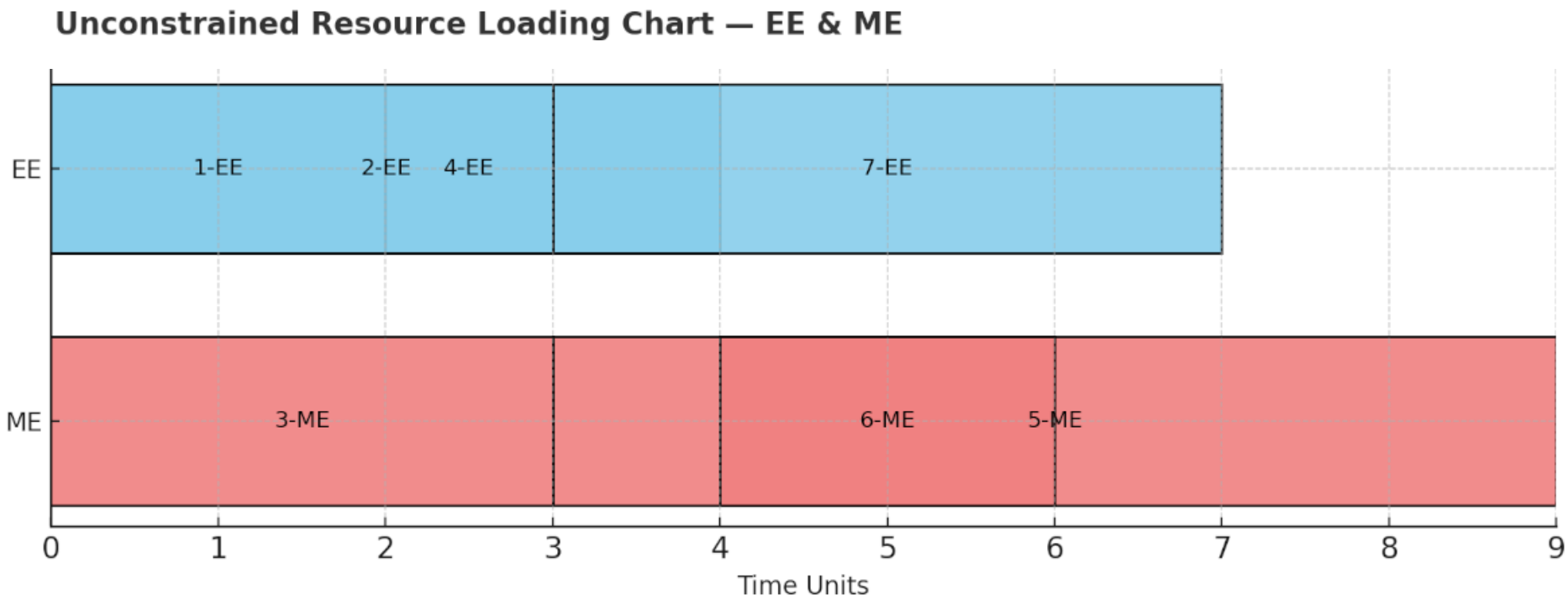
- **Project Duration: 9 days**
- **Critical Path:** The sequence of activities with zero slack is 1 → 4 → 5.
- **Critical Activities:** Activities 1, 4, and 5 have zero slack, making them critical.

- **Could something like this happen in real projects?**
- Yes, this situation happens in real projects. Resource limits often force changes in activity timing, increase total project duration, and alter which activities are critical — even when the precedence relationships remain the same.

Exercise 4 - Unconstrained Schedule



Exercise 4 - Unconstrained Schedule



Exercise 7 - Tractor Scheduling

Applying the Parallel Method

This schedule was created using the parallel method with three available tractors and the specified heuristics (min slack, smallest duration, lowest ID).

- **Final Project Duration: 14 Days**
- **Scheduling Notes:**
  - Activities C and F were delayed because not enough tractors were available when they were ready to start.
  - The heuristics ensured that critical activities (those with the least slack, like D and F) were scheduled first when resources were contested.

Exercise 7 - Tractor Scheduling

ID	RES	DUR	ES	LF	SL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	1	4	0	5	0	1	1	1	1												
B	2	5	0	5	0	2	2	2	2	2											
C	2	4	4	10	2					2	2	2	2								
D	1	5	5	10	0						1	1	1	1	1						
E	2	3	5	10	2						2	2	2								
F	2	2	10	12	0												2	2			
Resources scheduled						3	3	3	3	4	5	5	5	1	1		2	2			
Resources available						3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

ID	RES	DUR	ES	LF	SL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	1	4	0	5	0	1	1	1	1												
B	2	5	0	5	0	2	2	2	2	2											
C	2	4	4	10	2					X	X	X	X	2	2	2	2				
D	1	5	5	10	0						1	1	1	1	1						
E	2	3	5	10	2						2	2	2								
F	2	2	10	12	0												X	X	2	2	
Resources scheduled						3	3	3	3	2	3	3	3	3	3		2	2	2	2	
Resources available						3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3

Exercise 7 - Tractor Scheduling

Activity (ID)	Duration (DUR)	Early Start (ES)	Early Finish (EF=ES+DUR)	Late Start (LS=LF−DUR)	Late Finish (LF)	Slack (SL)
A	4	0	4	1	5	0
B	5	0	5	0	5	0
C	4	8	12	6	10	-3
D	5	5	10	5	10	0
E	3	5	8	7	10	2
F	2	12	14	10	12	-2

Exercise 10 - Carpenter Scheduling

Adapting to Fewer Resources

The project team was reduced from three carpenters to two, requiring a new resource-constrained schedule.

- **Final Project Duration:** 16 Days
- **Bonus Status:** The project will not receive the bonus, as completion takes longer than the 15-day target.
- **Impact:** The loss of one carpenter delayed the project significantly, showing the high impact of resource availability on a project timeline.
- **You will receive a bonus if the project is completed within 15 days. Should you start planning how you will spend your bonus?**
- No, we should not be planning to spend the bonus as the project will be completed after 15 days with an additional delay of 1 day

Exercise 10 - Carpenter Scheduling

ID	RES	DUR	ES	LF	SL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	2	2	0	2	0	2	2																	
B	2	1	2	5	2			2																
C	2	3	2	5	0			2	2	2														
D	2	1	5	6	0						2													
E	2	2	6	8	0							2	2											
F	1	3	6	10	1							1	1	1										
G	2	2	8	10	0										2	2								
H	2	2	10	12	0												2	2						
Resources scheduled						2	2	4	2	2	2	3	3	3	2	2	2							
Resources available						2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

ID	RES	DUR	ES	LF	SL	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
A	2	2	0	2	0	2	2																	
B	2	1	2	5	2				X	X	X	2												
C	2	3	2	5	0				2	2	2													
D	2	1	5	6	0							X	2											
E	2	2	6	8	0								X	2	2									
F	1	3	6	10	1								X	X	X	X	X	X	X	1	1	1		
G	2	2	8	10	0										X	2	2							
H	2	2	10	12	0												X	2	2					
Resources scheduled						2	2	2	2	2	2	2	2	2	2	2	2	2	2	1	1	1		
Resources available						2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2

Exercise 10 - Carpenter Scheduling

Activity (ID)	Duration (DUR)	Early Start (ES)	Early Finish (EF=ES+DUR)	Late Start (LS=LF−DUR)	Late Finish (LF)	Slack (SL)
A	2	0	2	0	2	0
B	1	5	6	4	5	-1
C	3	2	5	2	5	0
D	1	6	7	6	7	-1
E	2	7	9	6	8	-1
F	3	13	16	7	10	-6
G	2	9	11	8	10	-1
H	2	11	13	10	12	-1

Case 8.2 - The Power Train Ltd. Dilemma

The Problem: Scheduling Chaos

Power Train Ltd. cannot expand until its project scheduling conflicts are resolved. The current system lacks a formal process for assigning engineers, leading to delays and inefficiency.

The Solution: A Multi-Criteria Approach

Claude should implement a priority-based scheduling system.

- **Key Criteria for Decisions:**
  - Project Strategic Importance
  - Engineer Expertise and Fit
  - Project Complexity
  - Customer Importance
  - Resource Availability
  - Cost Efficiency

This approach ensures the right people are on the right projects at the right time.

Case 8.2 - The Power Train Ltd. Dilemma

Recommended sequence:

- Rank projects by strategic value
- Match skills to needs
- Check availability
- Apply scheduling heuristics
- Review cost efficiency
- Monitor via PMO

By following this systematic sequence, Power Train Ltd. can ensure that the right people are assigned to the right projects at the right time, improving efficiency, reducing conflicts, and positioning the company for successful expansion into the truck power train market.

Case A8.1 - The CCPM Dilemma: Introduction

Report on Project Failures in the IT Division

- **Background:** The IT division has struggled to complete projects on time, even after trying schedule padding and adopting Critical Chain Project Management (CCPM).
- **CCPM Failure:** The CCPM implementation was flawed, leading to unrealistic schedules, demoralized staff, and continued delays.
- **Objective:** This report identifies the root causes of failure and proposes a corrective action plan.

The CCPM Dilemma: Key Problems

Unrealistic Activity Estimates: Durations were cut far below the recommended 50% confidence level, making tasks impossible and creating distrust.

Inadequate Buffers: Project and feeding buffers were too small or omitted, leaving projects unprotected from delays. Management misunderstood buffers as inefficiency instead of strategic protection.

Excessive Multitasking: Employees were assigned to up to six concurrent projects, destroying focus and productivity.

Misunderstanding of CCPM: The methodology was used to simply reduce time, not to optimize flow and resources. Buffer consumption was not tracked to assess risk.

The CCPM Dilemma: Root Causes & Culture Issues

Lack of Training: Managers and staff were not properly trained in CCPM, leading to confusion.

Low Morale and Burnout: Unrealistic deadlines and workload caused high stress and employee turnover.

No Project Prioritization: All projects were treated as urgent, overloading the system.

Root Causes Summary:

- Misinterpretation of CCPM methodology.
- Overloaded resource capacity.
- Lack of team involvement in planning.
- Inadequate leadership training.

The CCPM Dilemma: Plan of Action (Part 1)

Rebuild Trust and Realistic Scheduling

- Reinstate collaborative estimation with teams using the 50% probability guideline.
- Communicate that the goal is accuracy and sustainability, not just speed.

Apply CCPM Correctly

- Identify the true critical chain (task and resource dependencies).
- Implement proper project and feeding buffers.
- Monitor progress with buffer burn charts.

Manage Resources & Reduce Multitasking

- Limit the number of concurrent projects per resource.
- Adopt a staggered project start policy—begin new projects only when capacity allows.

The CCPM Dilemma: Plan of Action (Part 2)

Introduce Project Prioritization

- Develop a project prioritization matrix based on strategic value.
- Create a Project Management Office (PMO) to oversee resource allocation.

Strength Training and Learning

- Conduct comprehensive CCPM workshops for all staff and managers.
- Establish regular post-project reviews to capture lessons learned.

Improve Communication and Culture

- Foster a transparent and collaborative culture where team input matters.
- Shift management's role from task controllers to facilitators and enablers.

Conclusion

Effective Scheduling requires balancing dependencies, resources, and strategic priorities.

Resource Constraints are a primary driver of project delays and must be proactively managed.

Methodologies like CCPM are powerful when implemented correctly with proper training and cultural buy-in.